WoSM Academy

CS102-F26 · DIGITAL DESIGN AND COMPUTER ARCHITECTURE I

Boolean Algebra

Lecture 03: the laws and theorems of an algebra of 0 and 1

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Today

01 An algebra of 0 and 1

02 The seven laws

03 Theorems

04 Simplifying expressions

05 Summary

01

An algebra of 0 and 1

From school algebra to Boolean algebra

School algebra works with numbers, variables and many operations:

$$3x^2 - \frac{4y}{z} + \frac{2}{7}xz^3, \qquad x, y, z \in \mathbb{R}.$$

In Boolean algebra we restrict both:

- the values to the set $\{0, 1\}$;
- the operations to addition (+), multiplication ($\cdot$) and negation ($'$).

Some laws are the same as in school, some are new. For example, with $x = y = 1$, school algebra gives $x + y = 2$. There is no 2 in $\{0, 1\}$: in Boolean algebra $1 + 1 = 1$.

Named after George Boole, The Laws of Thought (1854); see Lecture 01.

The three operations

a	b	$a + b$
0	0	0
0	1	1
1	0	1
1	1	1

OR

a	b	$a \cdot b$
0	0	0
0	1	0
1	0	0
1	1	1

AND

a	a'
0	1
1	0

NOT

- $a + b$ is 1 when *at least one* of a, b is 1.
- $a \cdot b$ is 1 only when *both* are 1.
- a' is the *complement* of a.

As logic gates, these are the OR, AND and NOT gates (Unit 6).

Notation and precedence

- The complement has three common notations: a', $\neg a$ and $\overline{a}$. These slides use a', as the book does.
- The dot is often left out: ab means $a \cdot b$.
- As in school, multiplication binds more tightly than addition, and the complement binds most tightly of all:

$$a + b'c \;=\; a + \big((b') \cdot c\big).$$

- To complement more than one variable, use brackets: $(ab)'$ is not $a'b'$.

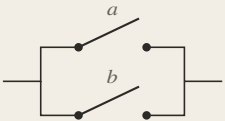
> **Warning**
>
> $(ab)'$ and $a'b'$ are different functions: for $a = 1$, $b = 0$ the first is 1 and the second is 0.

Switches: series and parallel

Claude Shannon (1937): Boolean algebra describes circuits of switches. Let 1 mean a closed switch, or current flowing.



Series: current flows only if *a and b* are closed: $a \cdot b$.



Parallel: current flows if *a or b* is closed: $a + b$.

Two switches in parallel, both closed, still carry current: $1 + 1 = 1$.

02

The seven laws

Laws 1–3

Definition (Commutativity)

The order does not matter: $a + b = b + a, \quad a \cdot b = b \cdot a.$

Definition (Associativity)

The grouping does not matter: $(a + b) + c = a + (b + c), \quad (ab)c = a(bc).$

Definition (Distributivity)

Expand or factor: $a(b + c) = ab + ac, \quad a + bc = (a + b)(a + c).$

Laws 4–7

Definition (Identity)

0 is the identity for +, and 1 for ·:

$$a + 0 = a, \quad a \cdot 1 = a$$

Definition (Idempotence)

An operand combined with itself:

$$a + a = a, \quad a \cdot a = a$$

Definition (Complement)

With its complement, the sum is 1 and the product is 0:

$$a + a' = 1, \quad a \cdot a' = 0$$

Definition (Domination)

Adding 1 gives 1; multiplying by 0 gives 0:

$$a + 1 = 1, \quad a \cdot 0 = 0$$

Not your school algebra

- $a + a = a$, not $2a$; $\quad a + 1 = 1$, not $a + 1$.
- The second distributive law, $a + bc = (a + b)(a + c)$, is new. With ordinary numbers it fails: $1 + 2 \cdot 3 = 7$, but $(1 + 2)(1 + 3) = 12$.
- There is no subtraction or division, so you cannot cancel: $a + b = a + c$ does not mean $b = c$. Try $a = 1$, $b = 0$, $c = 1$.

> **Remark**
>
> The laws are the axioms of Boolean algebra: we accept them as they are stated, and everything else is derived from them.

Duality

Look again at the laws: each one is a pair. The second equation comes from the first by swapping $+ \leftrightarrow \cdot$ and $0 \leftrightarrow 1$.

Law	Equation	Its dual
Identity	$a + 0 = a$	$a \cdot 1 = a$
Complement	$a + a' = 1$	$a \cdot a' = 0$
Distributivity	$a(b + c) = ab + ac$	$a + bc = (a + b)(a + c)$

> **The duality principle**
>
> If an equation follows from the laws, so does its dual. Every proof gives you a second theorem for free.

Checking a law by exhaustion

A variable has only two values, so we can try them all. For $a + bc = (a + b)(a + c)$:

a	b	c	bc	$a + bc$	$a + b$	$a + c$	$(a + b)(a + c)$
0	0	0	0	**0**	0	0	**0**
0	0	1	0	**0**	0	1	**0**
0	1	0	0	**0**	1	0	**0**
0	1	1	1	**1**	1	1	**1**
1	0	0	0	**1**	1	1	**1**
1	0	1	0	**1**	1	1	**1**
1	1	0	0	**1**	1	1	**1**
1	1	1	1	**1**	1	1	**1**

The two bold columns agree on every row, so the equation holds. With n variables there are 2^n rows (Lecture 02).

03

Theorems

Two ways to prove an equation

Proof by exhaustion

- Compute both sides for every combination of values.
- Always works, but needs 2^n rows: fine for 3 variables, not for 30.

Algebraic proof

- Turn one side into the other, one law per step.
- Short when you see the way; and each step shows *why*.

Example (Idempotence follows from the other laws)

$$
\begin{aligned}
a + a &= (a + a) \cdot 1 && \texttt{identity} \\
&= (a + a)(a + a') && \texttt{complement} \\
&= a + aa' && \texttt{distributivity} \\
&= a + 0 = a && \texttt{complement, identity}
\end{aligned}
$$

De Morgan's theorem

Theorem (De Morgan)

The complement of a sum is the product of the complements, and the complement of a product is the sum of the complements:

$$(a + b)' = a' \cdot b', \qquad (a \cdot b)' = a' + b'.$$

a	b	$(a+b)'$	$a' \cdot b'$	$(a \cdot b)'$	$a' + b'$
0	0	$0' = 1$	$1 \cdot 1 = 1$	$0' = 1$	$1 + 1 = 1$
0	1	$1' = 0$	$1 \cdot 0 = 0$	$0' = 1$	$1 + 0 = 1$
1	0	$1' = 0$	$0 \cdot 1 = 0$	$0' = 1$	$0 + 1 = 1$
1	1	$1' = 0$	$0 \cdot 0 = 0$	$1' = 0$	$0 + 0 = 0$

Using De Morgan

- It works for any number of variables: $(a + b + c)' = a'b'c'$ and $(abc)' = a' + b' + c'$.
- To complement an expression: swap + and ·, and complement every variable. Keep the original grouping with brackets.

Example (Complement of $ab + c$)

$$\begin{aligned}(ab + c)' &= (ab)' \cdot c' && \text{De Morgan} \\ &= (a' + b') \cdot c' && \text{De Morgan}\end{aligned}$$

Coming back: De Morgan's theorem is why a single kind of gate, NAND or NOR, can build every circuit (Unit 6).

Absorption

Theorem (Absorption)

$$a + a \cdot b = a, \qquad a \cdot (a + b) = a.$$

$a + ab = a \cdot 1 + a \cdot b$	identity
$= a \cdot (1 + b)$	distributivity
$= a \cdot 1$	domination
$= a$	identity

$a(a + b) = aa + ab$	distributivity
$= a + ab$	idempotence
$= a$	the left proof

In words: if a is 1, both sides are 1; if a is 0, both sides are 0. The value of b never matters.

Consensus

Theorem (Consensus)

$$a \cdot b + a' \cdot c + b \cdot c = a \cdot b + a' \cdot c$$

The term bc is redundant: whenever $b = c = 1$, either $a = 1$ and $ab = 1$, or $a = 0$ and $a'c = 1$.

$$\begin{aligned} ab + a'c + bc &= ab + a'c + bc(a + a') && \text{identity, complement} \\ &= ab + abc + a'c + a'bc && \text{distributivity, commutativity} \\ &= ab + a'c && \text{absorption, twice} \end{aligned}$$

Its dual: $(a + b)(a' + c)(b + c) = (a + b)(a' + c)$.

Consensus by exhaustion

a	b	c	ab	$a'c$	bc	$ab + a'c + bc$	$ab + a'c$
0	0	0	0	0	0	**0**	**0**
0	0	1	0	1	0	**1**	**1**
0	1	0	0	0	0	**0**	**0**
0	1	1	0	1	1	**1**	**1**
1	0	0	0	0	0	**0**	**0**
1	0	1	0	0	0	**0**	**0**
1	1	0	1	0	0	**1**	**1**
1	1	1	1	0	1	**1**	**1**

In both rows where $bc = 1$, another term is already 1, so dropping bc changes nothing.

Transposition

Theorem (Transposition)

$$(a+b)\cdot(a'+c) = a\cdot c + a'\cdot b$$

$$\begin{aligned}
(a+b)(a'+c) &= (a+b)\,a' + (a+b)\,c && \text{distributivity}\\
&= a'a + a'b + ac + bc && \text{distributivity, commutativity}\\
&= 0 + a'b + ac + bc && \text{complement}\\
&= ac + a'b + bc && \text{identity, commutativity}\\
&= ac + a'b && \text{consensus}
\end{aligned}$$

The last step is consensus with the roles of b and c swapped. The book shows it by case analysis on bc: if $bc = 1$ then $b = c = 1$, and $ac + a'b = a + a' = 1$ as well.

Involution

Theorem (Involution)

$$(a')' = a$$

a	$(a')'$
0	$(0')' = 1' = 0$
1	$(1')' = 0' = 1$

Complementing twice gives back what you started with: two NOT gates in a row change nothing (Unit 6).

All in one place

		Dual
Commutativity	$a + b = b + a$	$ab = ba$
Associativity	$(a + b) + c = a + (b + c)$	$(ab)c = a(bc)$
Distributivity	$a(b + c) = ab + ac$	$a + bc = (a + b)(a + c)$
Identity	$a + 0 = a$	$a \cdot 1 = a$
Idempotence	$a + a = a$	$aa = a$
Complement	$a + a' = 1$	$aa' = 0$
Domination	$a + 1 = 1$	$a \cdot 0 = 0$
De Morgan	$(a + b)' = a'b'$	$(ab)' = a' + b'$
Absorption	$a + ab = a$	$a(a + b) = a$
Consensus	$ab + a'c + bc = ab + a'c$	$(a + b)(a' + c)(b + c) = (a + b)(a' + c)$
Transposition	$(a + b)(a' + c) = ac + a'b$	
Involution	$(a')' = a$	

04

Simplifying expressions

Two quick ones

$$
\begin{aligned}
ab + ab' &= a(b + b') && \texttt{distributivity} \\
&= a \cdot 1 && \texttt{complement} \\
&= a && \texttt{identity}
\end{aligned}
\qquad
\begin{aligned}
(a + b)(a + b') &= a + bb' && \texttt{distributivity} \\
&= a + 0 && \texttt{complement} \\
&= a && \texttt{identity}
\end{aligned}
$$

The two are duals of each other. Whatever b is, the result depends only on a.

> **Coming back**
>
> Combining two terms that differ in one variable, like ab and ab', is the idea behind Karnaugh maps `(Unit 5)`.

A longer one

Simplify $f = a'b + ab + ab'$.

$$
\begin{aligned}
a'b + ab + ab' &= a'b + a(b + b') && \text{distributivity} \\
&= a'b + a \cdot 1 = a'b + a && \text{complement, identity} \\
&= a + a'b && \text{commutativity} \\
&= (a + a')(a + b) && \text{distributivity} \\
&= 1 \cdot (a + b) = a + b && \text{complement, identity}
\end{aligned}
$$

Three products and two sums became one sum. Check one row: $a = 0$, $b = 1$ gives $1 + 0 + 0 = 1$ on the left and $0 + 1 = 1$ on the right.

Why simplify?

- Each operation will become a logic gate (Unit 6). A simpler expression is a smaller, cheaper and often faster circuit.
- $a'b + ab + ab'$ needs two NOTs, three ANDs and two ORs; $a + b$ needs one OR.
- Algebra finds simplifications, but you have to see them. Karnaugh maps give a systematic way for small functions (Unit 5).

Try it

Simplify, naming the law for each step.

1. $a + a'b$
2. $a(a' + b)$
3. $(a' + b')'$

Try it

Simplify, naming the law for each step.

1. $a + a'b$

   Answer: $(a + a')(a + b) = 1 \cdot (a + b) = a + b$ (distributivity, complement, identity)

2. $a(a' + b)$

3. $(a' + b')'$

Try it

Simplify, naming the law for each step.

1. $a + a'b$
   Answer: $(a + a')(a + b) = 1 \cdot (a + b) = a + b$ (distributivity, complement, identity)
2. $a(a' + b)$
   Answer: $aa' + ab = 0 + ab = ab$ (distributivity, complement, identity)
3. $(a' + b')'$

Try it

Simplify, naming the law for each step.

1. $a + a'b$
   Answer: $(a + a')(a + b) = 1 \cdot (a + b) = a + b$ (distributivity, complement, identity)
2. $a(a' + b)$
   Answer: $aa' + ab = 0 + ab = ab$ (distributivity, complement, identity)
3. $(a' + b')'$
   Answer: $(a')'(b')' = ab$ (De Morgan, involution)

Try it

Simplify, naming the law for each step.

1. $a + a'b$
   Answer: $(a + a')(a + b) = 1 \cdot (a + b) = a + b$ (distributivity, complement, identity)
2. $a(a' + b)$
   Answer: $aa' + ab = 0 + ab = ab$ (distributivity, complement, identity)
3. $(a' + b')'$
   Answer: $(a')'(b')' = ab$ (De Morgan, involution)

> **Remark**
>
> The first two are a pair of duals; compare them with absorption.

05

Summary

Summary

- Boolean algebra: the values 0 and 1, and the operations + (OR), · (AND) and ′ (NOT).
- Seven laws: commutativity, associativity, distributivity, identity, idempotence, complement and domination. Each comes as a dual pair.
- Theorems: De Morgan, absorption, consensus, transposition and involution.
- Prove equations by exhaustion (2^n rows) or algebraically, one law per step.
- Simplifying an expression simplifies the circuit that computes it.

Before the next lecture

- Read Section 2.2 of the book, *Boolean Algebra*.
- Prove the dual of absorption, $a(a + b) = a$, by exhaustion.
- Prove the dual of consensus, $(a + b)(a' + c)(b + c) = (a + b)(a' + c)$, by the duality principle and by exhaustion.
- Simplify $abc + abc' + ab'$.

Next lecture (Unit 4)

Boolean functions: truth tables, and the canonical sum-of-products and product-of-sums forms.